

ZS-M40

Z-Spin Holonomy Knot Pairs and the Arithmetic Topology of Primes

V_XZ / V_ZY Half-Angle Doublet, $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ Arithmetic Frame, and the Sieve Shadow of (Z, X, Y) Sector Statistics

Author: Kenny Kang

Affiliation: Z-Spin Cosmology Collaboration

Date: May 2026

Theme / Paper Code: Math Spine — ZS-M40

Version: v1.1

VERIFICATION: 89/89 PASS | ANTI-NUMEROLOGY MC STRONG PASS (0.0098%) | ZERO FREE PARAMETERS

§0. Abstract

This paper introduces the J-conjugate holonomy knot pair $(K_p, \bar{K}_p)$ as the natural Z-sector realization of prime numbers in Z-Spin Cosmology. Building on three PROVEN corpus pillars — the half-angle channel pair $V_{XZ} \propto \exp(+i\theta/2)$, $V_{ZY} \propto \exp(-i\theta/2)$ with $V_{ZY} = (V_{XZ})^*$ (ZS-F4 §7/§7B), the 4π closure $D^{\wedge\{1/2\}}(4\pi) = +I$ on the $j = 1/2$ spinor module (ZS-M3 Lemma 10.1), and the Wilson-LOCATOR closed form $F_Q(p) = \sin(11\pi/p) / [11 \cdot \sin(\pi/p)]$ (ZS-M34 Theorem M34.V.6) — the prime p is identified with a pair of Wilson cycles carrying half-angle holonomy $\pm\theta/2$, prime period $4\pi/p$, $Q = 11$ colored framing $F_Q(p)$, and, when p is realized as an integer curvature in the M36 Apollonian Q-channel orbit, the mod-24 invariant $p \bmod 24 \in \{11, 23\}$ as an additional arithmetic-topological decoration.

Six principal theorems are established. M40.A (Cup Product $\leftrightarrow 4\pi$ Closure Compatibility Bridge) establishes a phase-class compatibility map between the $j = 1/2$ $SU(2)$ double-cover signature $\{+I, -I\}$ and the Tate–Mazur inv map $\text{inv} : H^3(\text{Spec}(O_K), \mathbb{G}_m) \simeq \mathbb{Q}/\mathbb{Z}$, sending $+I \mapsto 0$ and $-I \mapsto 1/2 \in \mathbb{Q}/\mathbb{Z}$; the $\pm I$ embedding into μ_2 is PROVEN, the Tate–Mazur isomorphism is IMPORTED-PROVEN, and their correspondence is registered as DERIVED-interpretation (pre-work 1: 43/43 PASS). M40.B (Multiplicative Gate $\leftrightarrow$ Galois Representation) identifies the corpus PROVEN operator M_p on $H_{\text{mult}} = \mathbb{C}[\mathbb{F}_{11}^{\times}]$ with the Galois representation $\text{Frob}_p \mapsto \text{diag}(\chi_k(p))$ on the character basis of $(\mathbb{Z}/11\mathbb{Z})^{\times}$ at DERIVED status (pre-work 2: 46/46 PASS), closing Gate FM13-5 from OPEN to DERIVED on the $M_p \leftrightarrow \text{Frob}_p$ identification; the further reduction of the Kim arithmetic CS functional to a Z-Spin Wilson cobordism action is separated and registered at HYPOTHESIS-strong. M40.C (J-Conjugate Holonomy Knot Pair Definition) establishes the principal new structural object of this paper. M40.D (Log-Holonomy Factorization) shows that integer prime factorization $N = \prod p_i^{\wedge\{a_i\}}$ is the additive log-image, via the HSI Theorem (ZS-M1), of the Z-sector holonomy decomposition $\ell_Z(K_N) = \sum a_i \cdot \ln p_i$. M40.E (Statistical Sieve Shadow) re-interprets the externally PROVEN prime-density statements of PNT, Mertens,

Dickman, and Sathe–Selberg as first-order shadows of K_p ensemble component statistics. M40.F (Anti-Numerology Uniqueness) records the pre-registered Monte Carlo result: $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ is the unique biquadratic imaginary field in the sample space $d \in [2, 1000]$ matching the Z-Spin V_4 conductor signature, and is uniquely class-number-one in this category.

An eight-fold structural bridge is documented between Z-Spin internal PROVEN objects and the externally PROVEN arithmetic Chern–Simons framework of Kim and collaborators (Kim 1510, 1609, 1706; Carlson–Kim 1911; Park–Park 2026), interpreting $\text{Spec}(O_K)$, the inv map, Galois representations, classical CS actions, n -th power residue symbols, the Cassels–Tate pairing, the Tate–Shafarevich group, and Kim 1510 §5’s speculative L-function path-integral construction in Z-Spin language. Bridge B8 supplies a cyclotomic operator-model analogue of Kim §5 at the specific level $Q = 11$ via the corpus PROVEN Euler product $\prod_{p \neq 11} \det(I - p^{-s} M_p)^{-1} = \zeta_{\mathbb{Q}(\zeta_{11})}(s)$ (ADS-2); the full path-integral closure in general is OPEN, with the externally PROVEN Carlson–Chung–D. Kim–M. Kim–Park–Yoo (2024) p -adic L-function framework identified as the proper closure anchor.

All inputs are LOCKED from upstream corpus: $A = 35/437$ (ZS-F2), $Q = 11$ (ZS-F5), $(Z, X, Y) = (2, 3, 6)$ (ZS-F5 / ZS-M19), $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ (ZS-M22), V_4 character decoration $(a_\chi, q_\chi) \in \{(0,1), (1,3), (1,11), (0,33)\}$ (ZS-M25). Zero new free parameters are introduced. Verification suites comprise pre-work 1 (cup product, 43/43 PASS), pre-work 2 ($M_p \leftrightarrow$ Galois, 46/46 PASS), and pre-work 3 / 3b (anti-numerology MC at $N = 500,000$, seed 35, plus extended robustness over $d \leq 1000$). Ten falsification gates F-M40.1 through F-M40.10 are registered.

Keywords: Z-Spin holonomy knot, J-conjugate half-angle pair, biquadratic field $\mathbb{Q}(\sqrt{-3}, \sqrt{-11})$, V_4 Galois group, Klein four-group, arithmetic Chern–Simons theory, multiplicative gate M_p , Wilson-LOCATOR, prime sieve shadow, Apollonian Q-channel, anti-numerology Monte Carlo, log-holonomy factorization.

Epistemic Status Legend

Table L.1. Epistemic status tags used throughout ZS-M40.

STATUS	DEFINITION
LOCKED	Core constant fixed in upstream paper; not adjustable downstream.
PROVEN	Formal proof from standard mathematics or exact symbolic derivation. Numerical agreement at machine or 50-digit precision is separately registered as VERIFIED.
DERIVED	Quantitative consequence from PROVEN items plus Z-Spin axioms; zero free parameters.
DERIVED-CONDITIONAL	Derived under explicitly stated upstream condition.
DERIVED-interpretation	Interpretive synthesis of PROVEN components; not a new theorem but a single-frame reading.

STATUS	DEFINITION
VERIFIED	Numerical or computational confirmation. For exhaustive finite-domain checks (e.g. enumeration over a declared finite range), the term COMPUTATIONALLY VERIFIED is preferred.
IMPORTED	Externally proved theorem used here without re-proof; full citation given. Equivalent to the longer tag IMPORTED-PROVEN .
HYPOTHESIS-strong	Multiple independent lines of structural support; explicit closure path identified.
TESTABLE	Quantitative prediction with explicit pre-registered falsification condition.
TESTABLE-LONG	Testable in principle but closure depends on long-horizon external developments.
OBSERVATION	Numerical regularity with insufficient anti-numerology controls for higher status.
NON-CLAIM	Explicit declaration of what this paper does NOT establish.
OPEN	Recognized gap with explicit closure path identified.

§1. Introduction — From M28 Dynamical Shadow to M40 Holonomy Knot

§1.1 Position in the Z-Spin program

The Z-Spin RH-Bridge program (ZS-M22 through ZS-M34) reads the corpus as the dynamical shadow of the analytic structure on the critical strip. ZS-M28 Theorem 28.5 catalogued the $\sigma = 1/2 \leftrightarrow j = 1/2$ structural isomorphism under a common Z_2 -involution. ZS-M33 Reading C routed all four V_4 -arithmetic mechanisms through the corpus-PROVEN Z-mediator projector $\Pi_Z = (1/2)(I + J_Z)$, with prime-specific Wilson-LOCATOR phase $\exp(2\pi i(j - 5)/p)$ as the non-separability witness. ZS-M34 v2.0 supplied the closed form $F_Q(p) = \sin(11\pi/p) / [11 \cdot \sin(\pi/p)]$ and the factorized indefinite kernel $\mathcal{K}_K = \mathcal{K}_{\text{arith}} \otimes \mathcal{K}_{\text{Wilson-Sonin}}$ (M34.V.8). ZS-M36 demonstrated that the Apollonian integer gasket with root quadruple $(-1, 2, 2, 3)$ hosts the prime-curvature Q-channel exactly at residues $\{11, 23\} \bmod 24$, verified for 19,413 primes.

This paper occupies a new structural slot in the program. ZS-M28 / M33 / M34 read the critical-strip side from the operator side. ZS-M36 read prime distribution through Apollonian arithmetic. ZS-M40 reads the prime itself — not the distribution, not the critical line, but the single integer p — as a structural object on the Z-sector. The reading is geometric: each prime p is identified with a pair of Wilson cycles carrying half-angle holonomy $\pm\theta/2$ on the channel pair (V_{XZ}, V_{ZY}) (ZS-F4 §7/§7B PROVEN), with prime period $4\pi/p$ and Q-colored framing $F_Q(p)$. For primes p realized as Apollonian curvatures in the M36 Q-channel, an additional mod-24 decoration $p \in \{11, 23\} \pmod{24}$ attaches; this is not a universal claim about all primes. The pair coexists at 90° phase-orthogonality per the Spin-Pair Coexistence Principle (Book PART X Reading X.4).

§1.2 What is new in M40

Three structural objects are new in this paper. First, the J-conjugate holonomy knot pair $(K_p, \bar{K}_p)$ is defined explicitly (Theorem M40.C, §3) as the assembly of six corpus PROVEN components into a single named object: V_{XZ} / V_{ZY} phase pair (F4), 4π closure (M3), Wilson-LOCATOR (M34), Apollonian Q-channel (M36), J-conjugation (F4 §7B), and Twin-Reuleaux geometric realization (S15). Second, the log-holonomy factorization $\ell_Z(K_N) = \sum a_i \cdot \ln p_i$ (Theorem M40.D, §4) is the direct image, under the HSI Theorem (M1), of the standard integer identity $\ln N = \sum a_i \cdot \ln p_i$; the contribution is the explicit Z-sector reading. Third, the eight-fold structural bridge to Kim arithmetic Chern–Simons (§8) explicitly identifies eight corpus PROVEN objects with eight externally PROVEN objects, with the principal interpretive contribution being the operator-level realization of Kim 1510 §5's speculative L-function path-integral construction by the corpus PROVEN Euler product $\prod_{p \neq 11} \det(I - p^{-s} M_p)^{-1} = \zeta_{\mathbb{Q}(\zeta_{11})}(s)$ (ADS-2).

The four empirical probability statements about prime density at scale 10^{100} — approximately 0.43% one-component, 2.36% two-component, 69.3% large-component dominance, 95.94% small-prime hit rate — are not new predictions of this paper. They are externally PROVEN statements of PNT, Sathe–Selberg, Dickman, and Mertens respectively. The contribution of ZS-M40 in §7 is to interpret them as first-order sieve shadows of the K_p ensemble component statistics; the statements themselves are IMPORTED, and §7 is registered as DERIVED-interpretation throughout.

§1.3 Anti-numerology discipline

The principal anti-numerology evidence in this paper is deterministic, not stochastic. Exhaustive enumeration over all distinct pairs (d_1, d_2) of squarefree positive integers in $[2, 1000]$ (183,921 candidate pairs) gives exactly one pair, $(3, 11)$, that matches the Z-Spin V_4 conductor signature $\{1, 3, 11, 33\}$ (Theorem M40.F, §9; COMPUTATIONALLY VERIFIED). Restricting further to pairs where both $\mathbb{Q}(\sqrt{-d_i})$ have class number $h = 1$, uniqueness is preserved: $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ is the unique biquadratic class-number-one field in this category. This is a literal uniqueness statement, not a statistical rarity claim. The pre-registered Monte Carlo (pre-work 3, seed = 35 = $A_{\text{numerator}} = 5 \times 7$, $N = 500,000$) supplies a secondary stochastic confirmation: the matching rate is 0.0098%, more than fifty times stricter than the pre-registered PASS threshold of 0.5%, and the background rate for generic biquadratic fields with composite third subfield (approximately 24%) makes the Z-Spin signature approximately 2,430 times rarer than background. The MC role is to confirm that sampling-with-replacement on $[2, 200]$ also yields only $(3, 11)$; the deterministic $[2, 1000]$ enumeration carries the actual uniqueness claim. These results are recorded in §9 (Theorem M40.F).

§1.4 Paper organization

§2 fixes notation and lists all LOCKED inputs. §3 establishes Theorem M40.C, the principal definition of $(K_p, \bar{K}_p)$. §4 derives the log-holonomy factorization (M40.D). §5 reports Theorem M40.A and pre-work 1 (cup product $\leftrightarrow 4\pi$ closure). §6 reports Theorem M40.B and pre-work 2 ($M_p \leftrightarrow$ Galois representation). §7 records the statistical sieve shadow interpretation (M40.E). §8 documents the

eight-fold Kim bridge. §9 records the anti-numerology Monte Carlo (M40.F). §10 places M40 in the M28 / M33 / M36 program. §11 lists ten falsification gates. §12 records NON-CLAIMs. §13 concludes. Appendices A–C give detailed verification suite results.

§2. Locked Inputs

All quantities used in this paper are inherited unchanged from prior corpus papers. Zero new free parameters are introduced. Throughout the paper the symbols A , Q , K , V_{XZ} , V_{ZY} , J_Z , Π_Z , M_p , W_p , $F_Q(p)$ carry their corpus definitions.

Table 2.1. LOCKED inputs from upstream Z-Spin papers.

Quantity	Value / Statement	Source	Status
A (geometric impedance)	$35/437 = 0.080092$	ZS-F2 v1.0	LOCKED
Q (register dimension)	11 (prime)	ZS-F5 v1.0	PROVEN
(Z, X, Y) sector dims	$(2, 3, 6)$; $Z + X + Y = 11$	ZS-F5 / ZS-M19	PROVEN
(δ_X, δ_Y)	$(5/19, 7/23)$	ZS-F2 §4.2	PROVEN
K (composite field)	$\mathbb{Q}(\sqrt{-3}, \sqrt{-11})$; V_4 Galois	ZS-M22 §2.3	PROVEN
$\text{disc}(K)$	$1089 = 33^2$	ZS-M22 §7.2	PROVEN
V_4 characters	$\{1, \chi_{-3}, \chi_{-11}, \chi_{33}\}$	ZS-M22 §2.3	PROVEN
V_4 conductors q_χ	$\{1, 3, 11, 33\}$	ZS-M25 §6.3	PROVEN
V_4 parities a_χ	$\{0, 1, 1, 0\}$	ZS-M25 §6.3 / M27 §4	PROVEN
z^* (i-tetration FP)	$0.4383 + 0.3606 i$	ZS-M1 v1.0	PROVEN
V_{XZ} phase	$\exp(+i\theta/2)$, $\theta = \pi(1 - \varepsilon)$	ZS-F4 §7	DERIVED
V_{ZY} phase	$\exp(-i\theta/2) = (V_{XZ})^*$	ZS-F4 §7B	DERIVED
4π closure	$D^{\{1/2\}}(4\pi) = +I$, $D^{\{1/2\}}(2\pi) = -I$	ZS-M3 Lemma 10.1	PROVEN
$F_Q(p)$	$\sin(11\pi/p) / [11 \cdot \sin(\pi/p)]$	ZS-M34 Thm M34.V.6	PROVEN
W_p (transfer)	$\text{diag}(\exp(2\pi i(j - 5)/p))$	ZS-M4 Eq. 7	PROVEN
$W_p _Z$	$\exp(-i \sigma_Z \cdot 2\pi/p)$	ZS-M4 + ZS-M3 §10	PROVEN
M_p (mult. gate)	$M_p a\rangle = pa \bmod 11\rangle$ on $\mathbb{F}_{11}^*$	ZS-M (ADS-1)	PROVEN
M_p eigenvalues	$M_p \chi_k\rangle = \chi_k(p) \chi_k\rangle$	ZS-M (ADS-1) Thm	PROVEN
Euler product ($Q=11$)	$\prod_{p \neq 11} \det(I - p^{-s} M_p)^{-1} = \zeta_{\mathbb{Q}(\zeta_{11})}(s)$	ZS-M (ADS-2)	PROVEN
Apollonian root	$(-1, 2, 2, 3)$	ZS-M36 Thm M36.2	PROVEN
Q -channel mod 24	prime curvatures $\equiv \{11, 23\} \bmod 24$	ZS-M36 Thm M36.7	DERIVED

Quantity	Value / Statement	Source	Status
Π_Z projector	$(1/2)(I + J_Z)$	ZS-F0 §8.6 / ZS-M31 §4.0	PROVEN

§2.2 IMPORTED external theorems

Six external theorems are imported as IMPORTED-PROVEN (used without re-proof). Full citations in §References.

Source	Statement
PNT	Hadamard 1896, de la Vallée Poussin 1896: $\pi(x) \sim x / \ln x$.
Mertens	Mertens 1874/1898: $\prod_{p \leq x} (1 - 1/p) \sim e^{-\gamma} / \ln x$.
Dickman	Dickman 1930: $\rho(u)$ is the limiting probability that a random integer $\leq x$ has its largest prime factor $\leq x^{1/u}$; $\rho(2) = 1 - \ln 2$.
Sathe–Selberg	Sathe 1953, Selberg 1954: distribution of $\omega(n)$, the number of distinct prime factors.
Tate–Mazur	Tate 1962 + Mazur 1973: $\text{inv} : H^3(\text{Spec}(O_F), \mathbb{G}_m) \simeq \mathbb{Q}/\mathbb{Z}$ for F totally imaginary.
Kim 1510 / 1609	Arithmetic CS functional $\text{CS}_c(\rho)$ on Galois representation $\rho : \pi_1^{\text{ét}}(\text{Spec}(O_F)) \rightarrow G$.

§3. Theorem M40.C — J-Conjugate Holonomy Knot Pair Definition

§3.1 Statement

For each prime p we define the Z-Spin holonomy knot pair $(K_p, \bar{K}_p)$ as the pair of Wilson cycles on the Z-sector channel pair (V_{XZ}, V_{ZY}) carrying:

- (i) half-angle holonomy $\pm\theta/2$, with $\theta = \pi(1 - \varepsilon(r))$;
- (ii) prime period $4\pi/p$ under the corpus PROVEN $W_p|_Z = \exp(-i \sigma_Z \cdot 2\pi/p)$;
- (iii) Q-colored framing $F_Q(p) = \sin(11\pi/p) / [11 \cdot \sin(\pi/p)]$ (M34.V.6 PROVEN);
- (iv) Apollonian Q-channel arithmetic decoration: for primes $p > 3$ realized as curvatures in the M36 integer Apollonian gasket with root quadruple $(-1, 2, 2, 3)$, the residue $p \bmod 24 \in \{11, 23\}$ is attached as an additional arithmetic-topological invariant (M36.7 PROVEN). This is NOT a universal statement about all primes — it applies only to those primes appearing in the Apollonian Q-channel orbit;
- (v) J-conjugate doublet structure $\bar{K}_p = (K_p)^*$ via $V_{ZY} = (V_{XZ})^*$ (F4 §7B PROVEN);
- (vi) Twin-Reuleaux geometric realization $R_1 \leftrightarrow K_p, R_2 \leftrightarrow \bar{K}_p$ (S15 Thm S15.1 DERIVED).

The pair coexists at 90° phase-orthogonality per the Spin-Pair Coexistence Principle (Book PART X Reading X.4, DERIVED-interpretation strong), with phase difference $\arg(V_XZ) - \arg(V_ZY) = \theta$ a gauge-invariant observable.

§3.2 Derivation

The definition assembles six corpus PROVEN components into a single named object. Each component is independently established; the contribution of M40.C is the explicit identification of the prime-indexed assembly as the natural Z-sector structural object representing the integer p.

Step 1 (V_XZ, V_ZY phase pair, PROVEN, F4 §7/§7B). The half-angle holonomy structure

$$V_XZ(r) = \sqrt{A \cdot \varepsilon(r)} / \sqrt{(1 + A \varepsilon^2(r))} \cdot \exp(+i \theta(r)/2),$$

$$V_ZY(r) = (V_XZ(r))^* = \sqrt{A \cdot \varepsilon(r)} / \sqrt{(1 + A \varepsilon^2(r))} \cdot \exp(-i \theta(r)/2)$$

is established via three independent paths (O(1,1) spinor representation, U(1) half-holonomy, square-root of T_XY) all converging on the same expression. Numerical verification at 80 lattice points in $\varepsilon \in (0, 1)$ gives $\max |\text{Im}(V_ZY \cdot V_XZ)| = 0$.

Step 2 (Twin-Reuleaux realization, DERIVED, S15 §2). The Frobenius theorem applied to $\dim(Z) = 2$ forces the Z-sector state space to carry the complex structure of $\mathbb{C}$. The unique 2D associative division algebra is $\mathbb{C}$. The twin-Reuleaux pair (R_1, R_2) realizes the J-conjugate doublet on plane curves: $R_1 \leftrightarrow V_XZ$ (sequential closure, time-point), $R_2 \leftrightarrow V_ZY$ (parallel closure, space-point), $R_2 = J(R_1)$ under the J-involution $\theta \rightarrow \theta + \pi$.

Step 3 (4π closure, PROVEN, M3 Lemma 10.1). The $j = 1/2$ representation satisfies $D^{\{1/2\}}(2\pi) = -I$ and $D^{\{1/2\}}(4\pi) = +I$ (SU(2) center action). This is the defining algebraic signature of a spinor versus a vector representation of SO(3). The 4π period is therefore the topological period of K_p and $\bar{K}_p$ individually.

Step 4 (Prime period $4\pi/p$, PROVEN, M4 + M3 §10). The transfer operator $W_p = \text{diag}(\exp(2\pi i(j - 5)/p))$ restricted to the Z-sector takes the σ_z -rotation form on the $\{|4\rangle, |6\rangle\}$ basis,

$$W_p|_Z = \exp(-i \sigma_z \cdot 2\pi/p),$$

which implements a rotation by $4\pi/p$ per prime p (since the spinor $2\pi = \text{SU}(2)$ half-period). This is the Z-Spin holonomy mechanism by which each prime acts as a quantized rotation on the Z-sector.

Step 5 (Q-colored framing $F_Q(p)$, PROVEN, M34.V.6). The Wilson-LOCATOR closed form

$$F_Q(p) = \sin(11\pi/p) / [11 \cdot \sin(\pi/p)]$$

is verified at 50-digit mpmath precision against the direct sum $(1/11) \sum_{j=0}^{10} \exp(2\pi i(j-5)/p)$ with maximum error 1.39×10^{-16} . $F_Q(11) = 0$ (ramified, $\sin \pi = 0$); for $p \neq 11$, $|F_Q(p)| < 1$ with $F_Q(2) = -1/11$ the smallest case.

Step 6 (Apollonian Q-channel, DERIVED, M36.7). For prime $p > 3$ appearing as a curvature in the integer Apollonian gasket with root quadruple $(-1, 2, 2, 3)$ — the unique Z-Spin Type root by D_2 uniqueness (Sarnak) plus Forcing Theorem (M19) — one has $p \equiv 11$ or $23 \pmod{24}$. Equivalently $p \equiv -1 \pmod{12}$. Verified on 19,413 primes via depth-18 BFS to $|k| \leq 10^6$: 100% in $\{11, 23\} \pmod{24}$, zero counterexamples. The mod-24 invariant is doubled relative to mod-12 by ZS-M3 Lemma 10.1 ($D^{1/2}(2\pi) = -I$), realizing the bosonic $SO(3)$ / fermionic $SU(2)$ two-faces structure of F18 §6.6.

§3.3 Scope notes for M40.C

M40.C asserts the named object $(K_p, \bar{K}_p)$ as a structural assembly. The isotopy class of K_p in any standard external knot category (Reidemeister, finite-type, HOMFLY, Khovanov) is registered as OPEN-M40.knot; external knot-invariant matching is a closure path identified for future work. The corpus content is exhausted by (i)–(vi) above and is internally self-consistent.

[STATUS: DERIVED] Six corpus PROVEN/DERIVED inputs assembled into a single named object. Zero new free parameters.

§4. Theorem M40.D — Log-Holonomy Factorization

§4.1 Statement

For any positive integer N with unique prime factorization $N = \prod_i p_i^{a_i}$, define the Z-Spin holonomy length on the knot pair $(K_p, \bar{K}_p)$ by

$$\ell_Z(K_p) := \ln p.$$

Then the holonomy length of the composite knot K_N satisfies

$$\ell_Z(K_N) = \sum_i a_i \cdot \ell_Z(K_{p_i}) = \sum_i a_i \cdot \ln p_i = \ln N.$$

Equivalently, the integer prime factorization is the additive log-image of the Z-sector multiplicative holonomy decomposition.

§4.2 Derivation

The derivation is a direct image of the HSI Theorem (ZS-M1 Theorem 1.1 PROVEN) Step 3:

(3a) Z-sector states compose additively: the state $z = \ln \Omega + i\theta$ is the log-conformal factor, and Weyl rescalings compose multiplicatively ($\Omega_{\text{total}} = \Omega_1 \cdot \Omega_2$), so their logarithms add: $z_{12} = z_1 + z_2$.

(3b) Holonomies compose multiplicatively: the holonomy of a composite parallel transport is the product of individual holonomies, $H(\gamma_1 \circ \gamma_2) = H(\gamma_1) \cdot H(\gamma_2)$ (Kobayashi–Nomizu Ch. II).

Combining: under the HSI transfer map $T(z) = i^z = \exp((i\pi/2) z)$, the additive sum of Z-sector states $z_p := \ln p$ (one per prime factor) maps to the product of holonomies

$$T(\sum_i a_i \cdot z_{\{p_i\}}) = T(\ln N) = \exp((i\pi/2) \ln N) = \prod_i T(z_{\{p_i\}})^{a_i}.$$

In the Z-sector this is the statement that the composite holonomy K_N decomposes multiplicatively as $\prod_i K_{\{p_i\}}^{a_i}$, while its length (in the natural $\ln$ -additive metric on Z-sector states) decomposes additively as $\sum_i a_i \cdot \ln p_i = \ln N$. This is M40.D.

§4.3 Numerical anchor — the four standard prime-density statements

M40.D situates four externally PROVEN prime-density statements as the natural metric statements on the Z-knot ensemble. None of these statements is new in this paper; the contribution is the Z-sector reading. The four statements at scale $N \sim 10^{\{100\}}$ (taken as a canonical comparison scale):

Table 4.1. Standard prime-density statements at scale $N \sim 10^{\{100\}}$ and their ZS-M40 Z-knot reading.

Statement	External PROVEN	Numerical value	ZS-M40 Z-knot reading
100-digit integer is prime	PNT: $1/\ln N$	$1/\ln(10^{\{100\}}) \approx 0.434\%$	One-component K_p probability
100-digit integer has $\omega(n) = 2$	Sathe–Selberg	$\ln \ln N / \ln N \approx 2.36\%$	Two-component $K_{\{p,q\}}$ sector
100-digit integer has largest prime factor $> 10^{\{50\}}$	Dickman	$P(P(n) > \sqrt{N}) = \ln 2 \approx 0.6931$	Dominant K_p component fraction
100-digit integer has a prime factor $\leq 10^6$	Mertens	$1 - e^{-\gamma} / \ln(10^6) \approx 95.94\%$	Short-prime sieve completeness

[STATUS: DERIVED] The arithmetic identity $\ln N = \sum a_i \ln p_i$ is elementary; the Z-sector holonomy reading is the direct image under HSI (M1 PROVEN). Table 4.1 is registered as DERIVED-interpretation, with all numerical values IMPORTED from external PROVEN sources.

§5. Theorem M40.A — Cup Product $\leftrightarrow 4\pi$ Closure Compatibility Bridge

§5.1 Statement

The Z-Spin 4π closure $D^{\wedge\{1/2\}}(4\pi) = +I$ on the $j = 1/2$ spinor module of the Z-sector and the Tate–Mazur inv map $\text{inv} : H^3(\text{Spec}(O_K), \mathbb{G}_m) \simeq \mathbb{Q}/\mathbb{Z}$ are compatible under the natural $\mu_2 \hookrightarrow \mathbb{Q}/\mathbb{Z}$ phase-class embedding, in the sense that $+I \mapsto 0 \in \mathbb{Q}/\mathbb{Z}$ and the 2π closure $D^{\wedge\{1/2\}}(2\pi) = -I \mapsto 1/2 \in \mathbb{Q}/\mathbb{Z}$ (the Tate self-dual point and unique order-2 element). This is a compatibility bridge, not a claim of explicit isomorphism between the $SU(2)$ double-cover and the étale cohomology group H^3 ; the $\pm I$ embedding into μ_2 is PROVEN, the Tate–Mazur isomorphism is IMPORTED-PROVEN, and the phase-class correspondence is DERIVED-interpretation.

§5.2 Derivation

The bridge proceeds at the level of finite Galois cohomology, using only PROVEN corpus inputs and IMPORTED external theorems.

Step 1 (PROVEN, M3 Lemma 10.1). $D^{\wedge\{1/2\}}(2\pi) = -I$, $D^{\wedge\{1/2\}}(4\pi) = +I$ are the $SU(2)$ center action signatures.

Step 2 (IMPORTED, Tate 1962 + Mazur 1973). For F a totally imaginary number field, $\text{inv} : H^3(\text{Spec}(O_F), \mathbb{G}_m) \simeq \mathbb{Q}/\mathbb{Z}$ is an isomorphism. $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ is totally imaginary with signature $(r_1, r_2) = (0, 2)$.

Step 3 (PROVEN by combinatorics + numerical enumeration, pre-work 1). Under the natural correspondence (i) $\pm I \in \{\pm 1\} \subset \mu_2 \subset \mathbb{G}_m$, (ii) $\mu_2 \hookrightarrow \mathbb{Q}/\mathbb{Z}$ via $x \rightarrow \arg(x)/(2\pi) \bmod 1$, the algebraic signature $D^{\wedge\{1/2\}}(2\pi) = -I$ maps to the unique order-2 element $1/2 \in \mathbb{Q}/\mathbb{Z}$, and $D^{\wedge\{1/2\}}(4\pi) = +I$ maps to the identity $0 \in \mathbb{Q}/\mathbb{Z}$.

Step 4 (DERIVED, pre-work 1 Categories E–G). The cup product $H^1(K, \mu_n) \times H^1(K, \mu_n) \rightarrow H^2(K, \mu_n \otimes \mu_n)$ followed by $H^2 \times H^1 \rightarrow H^3$ is constructed using explicit cocycle representatives on the V_4 -graded module M_4 over O_K with conductor decoration $(1, 3, 11, 33)$. The 4π -monodromy class evaluates to $0 \in \mathbb{Q}/\mathbb{Z}$ under inv , in agreement with Step 3.

§5.3 Verification (pre-work 1)

Table 5.1. Pre-work 1 verification suite categories (43/43 PASS).

Cat.	Scope	Sample tests	Pass
A	V_4 character data LOCKED from M22 / M25	characters $\{1, \chi_{\{-3\}}, \chi_{\{-11\}}, \chi_{\{33\}}\}$; conductors $\{1, 3, 11, 33\}$; parities $\{0, 1, 1, 0\}$; $\chi_{\{33\}} = \chi_{\{-3\}} \cdot \chi_{\{-11\}}$	7/7
B	Legendre symbol values at small primes	(d/p) for $d \in \{-3, -11, 33\}$ and primes $p \leq 31$; ramified primes $p \in \{3, 11\}$ excluded	12/12

Cat.	Scope	Sample tests	Pass
C	μ_n embedding in $\mathbb{G}_m$	$+1 \rightarrow 0, -1 \rightarrow 1/2$ in $\mathbb{Q}/\mathbb{Z}$; order-2 element uniqueness	5/5
D	V_4 -graded cocycle construction	explicit 1-cocycles on V_4 characters; coboundary check	6/6
E	Cup product $H^1 \times H^1 \rightarrow H^2$ on group cochains	cocycle conditions for H^2 ; orthogonality of V_4 characters	5/5
F	$H^2 \times H^1 \rightarrow H^3$ via Tate self-duality	Tate pairing structural check; reduction to $\mathbb{Q}/\mathbb{Z}$	4/4
G	4π closure cohomology class equals 0 in $\mathbb{Q}/\mathbb{Z}$	monodromy class evaluation; Legendre check 13 mod 11 corrected (13 is QNR mod 11)	4/4

[STATUS: DERIVED-interpretation] 43/43 PASS. The $\pm 1 \rightarrow \{0, 1/2\}$ embedding into $\mathbb{Q}/\mathbb{Z}$ is PROVEN; the Tate–Mazur sheaf-theoretic isomorphism $\text{inv} : H^3(\text{Spec}(O_K), \mathbb{G}_m) \simeq \mathbb{Q}/\mathbb{Z}$ is IMPORTED-PROVEN (Tate 1962 + Mazur 1973 + Kim 1510 Eq. (1)); their phase-class correspondence is DERIVED-interpretation. Pre-work 1 verifies the cup-product cohomology bookkeeping (cocycle conditions, Tate self-duality, V_4 -graded module M_4) underlying the bridge but does not claim a new external étale-cohomology theorem.

§6. Theorem M40.B — $M_p \leftrightarrow$ Galois Representation

§6.1 Statement

The Z-Spin Multiplicative Gate M_p on $H_{\text{mult}} = \mathbb{C}[\mathbb{F}_{11}^\times]$ (ZS-M corpus PROVEN ADS-1) is the Galois representation

$$\rho_p : \text{Frob}_p \mapsto \text{diag}(\chi_k(p)) \text{ on the character basis } \{|\chi_k\rangle : k = 0, \dots, 9\}$$

of $(\mathbb{Z}/11\mathbb{Z})^\times$. This operator identification $M_p = \rho_p$ closes Gate FM13-5 from OPEN to DERIVED on the $M_p \leftrightarrow \text{Frob}_p$ side; we separately register, at HYPOTHESIS-strong, the further claim that the Kim arithmetic Chern–Simons functional $\text{CS}_c(p)$ on $\text{Spec}(O_{\{\mathbb{Q}(\zeta_{11})\}})$ reduces, under the Kummer correspondence, to a Z-Spin Wilson cobordism action. The latter has not been written in operator-level explicit form in this paper and is OPEN-M40.B4.

§6.2 Derivation

M_p is defined as the permutation matrix $M_p|a\rangle = |pa \bmod 11\rangle$ for $a \in \mathbb{F}_{11}^\times$ and prime $p \neq 11$. Since $\gcd(p, 11) = 1$, M_p is unitary. By ADS-1 (M22 §3.2 PROVEN), M_p is diagonal in the Dirichlet character basis with eigenvalue $\chi_k(p)$.

The standard Galois action of Frob_p on the cyclotomic field $\mathbb{Q}(\zeta_{11})$ is given by $\text{Frob}_p : \zeta_{11} \rightarrow \zeta_{11}^p$. On the character basis χ_k of $\text{Gal}(\mathbb{Q}(\zeta_{11})/\mathbb{Q}) \cong (\mathbb{Z}/11\mathbb{Z})^\times$, Frob_p acts by $\chi_k(p)$. Therefore the eigenvalue assignments of M_p and $\rho(\text{Frob}_p)$ coincide, and the operator identification $M_p = \rho_p$ is established.

The Euler product identity (ADS-2, PROVEN, ZS-M22 §3.2):

$$\prod_{\{p \neq 11\}} \det(\mathbf{I} - p^{-s} \mathbf{M}_p)^{-1} = \prod_{\chi \bmod 11} L(s, \chi) = \zeta_{\mathbb{Q}(\zeta_{11})}(s)$$

realizes the cyclotomic Dedekind zeta factorization at the operator level. Numerical verification at $s = 2$ with primes $p \leq 200$ gives relative error 5.4×10^{-16} .

§6.3 Closing Gate FM13-5

Gate FM13-5 (Standard Artin L-function realization, OPEN since ZS-M13) asked whether the additive-character side $W_p = \text{diag}(\exp(2\pi i j / p))$ and the multiplicative-character side $\rho_p = \text{diag}(\chi_k(p))$ connect into a single operator-level Artin L-function. M40.B identifies the connection: W_p sits on the additive (spectral) side of $(\mathbb{Z}/p\mathbb{Z}, +)$; M_p sits on the multiplicative (Galois) side of $(\mathbb{Z}/11\mathbb{Z})^\times$; the explicit operator identification of M_p with ρ_p via ADS-1 closes the gap. Gate FM13-5: OPEN $\rightarrow$ DERIVED.

§6.4 Verification (pre-work 2)

Table 6.1. Pre-work 2 verification suite categories (46/46 PASS).

Cat.	Scope	Sample tests	Pass
A	$\mathbb{F}_{11}^\times$ structure and primitive root	$\mathbb{F}_{11}^\times$ has order 10; $g = 2$ is primitive root; cycle $\{1, 2, 4, 8, 5, 10, 9, 7, 3, 6\}$	5/5
B	M_p permutation matrix construction	$M_2, M_3, M_5, M_7, M_{13}$ explicit; $M_p \cdot M_p^{-1} = I$	6/6
C	Dirichlet character basis diagonalization (ADS-1)	$M_p \chi_k\rangle = \chi_k(p) \chi_k\rangle$; max eigen-error $< 10^{-10}$	10/10
D	$M_p \leftrightarrow \rho(\text{Frob}_p)$ on $(\mathbb{Z}/11\mathbb{Z})^\times$	$\text{Frob}_p : \zeta \rightarrow \zeta^p$; character eigenvalues match	5/5
E	Euler product (ADS-2) at $s = 2$	$\prod \det(\mathbf{I} - p^{-s} \mathbf{M}_p)^{-1} = \zeta_{\mathbb{Q}(\zeta_{11})}(s)$; rel.err 5.4×10^{-16}	4/4
I	Dirichlet class number formula $L(1, \chi_{-11})$	$L(1, \chi_{-11}) = \pi/\sqrt{11} \approx 0.9472$ ($h(-11) = 1$, $w = 2$)	4/4
J	Q-channel mod 24 density	primes $\equiv \{11, 23\} \bmod 24$; sample density $24.85\% \approx$ theoretical 25%	6/6
D.3 + J.8	Spectrum sort stability + O-M27.4 closure	ramified primes excluded; unramified curvatures carry $\sigma_3 = [S_2]$	6/6

[STATUS: DERIVED] 46/46 PASS. Gate FM13-5: OPEN $\rightarrow$ DERIVED.

§7. Theorem M40.E — Statistical Sieve Shadow Re-interpretation

§7.1 Statement

The four standard prime-density statements of analytic number theory at scale $N \sim 10^{\{100\}}$ (PNT, Sathe–Selberg, Dickman, Mertens) are first-order sieve shadows of the K_p ensemble component statistics under the M40.C definition and M40.D log-holonomy factorization.

§7.2 The four statements as Z-knot statistics

Each statement is externally PROVEN. The Z-sector reading is given alongside, marked DERIVED-interpretation. No new prime-distribution prediction is made by this paper; the contribution is the explicit Z-knot reading of the four statements.

(i) PNT (one-component K_p). The probability that a random integer near N is prime is, by the Prime Number Theorem (Hadamard 1896; de la Vallée Poussin 1896, IMPORTED-PROVEN), asymptotic to $1/\ln N$. At $N = 10^{\{100\}}$ this is $1/(100 \cdot \ln 10) \approx 0.434\%$. In M40 language: a random 100-digit integer has approximately a 0.4% chance of being a one-component holonomy knot K_p (no nontrivial composite knot decomposition under M40.D).

(ii) Sathe–Selberg (two-component $K_{\{p,q\}}$). The distribution of $\omega(n)$, the number of distinct prime divisors of n , is given by the Sathe–Landau–Selberg theorem (Sathe 1953; Selberg 1954, IMPORTED-PROVEN): $P(\omega(n) = k) \sim (\ln \ln N)^{\{k-1\}} / ((k-1)! \cdot \ln N)$ for fixed k as $N \rightarrow \infty$. For $k = 2$ this reduces to $\ln \ln N / \ln N$, which evaluates to $5.439 / 230.259 \approx 2.36\%$ at $N = 10^{\{100\}}$ (using $\ln 10^{\{100\}} = 230.2585$, $\ln \ln 10^{\{100\}} = 5.4391$). In M40 language: approximately 2.36% of 100-digit integers admit a two-component knot decomposition $K_N = K_{\{p_1\}} \sqcup K_{\{p_2\}}$.

(iii) Dickman (dominant K_p component). The Dickman function $\rho(u)$ gives the limiting probability that the largest prime factor $P^*(n)$ of a random integer $n \leq x$ is at most $x^{\{1/u\}}$. By Dickman (1930, IMPORTED-PROVEN), $\rho(2) = 1 - \ln 2 \approx 0.30685$. The complementary statement, which is the relevant one for the Z-knot reading, is that the largest prime factor exceeds $\sqrt{N}$ with probability $1 - \rho(2) = \ln 2 \approx 0.69315$. In M40 language: approximately 69.3% of integers near $10^{\{100\}}$ admit a single dominant K_p component carrying more than half the total log-holonomy length $\ell_Z(K_N) = \ln N$.

(iv) Mertens (short-prime sieve). The Mertens product theorem (Mertens 1874, IMPORTED-PROVEN) gives $\prod_{p \leq x} (1 - 1/p) \sim e^{\{-\gamma\}} / \ln x$ as $x \rightarrow \infty$. The probability that a random integer has no prime factor $\leq x$ is therefore asymptotic to $e^{\{-\gamma\}} / \ln x$, independent of N (provided $N \gg x$ is taken sufficiently large to average over). For $x = 10^6$: $e^{\{-\gamma\}} / \ln 10^6 = 0.56146 / 13.8155 = 0.04064$, hence the has-factor probability is $1 - 0.04064 \approx 95.94\%$. In M40 language: approximately 95.94% of 100-digit integers admit at least one short K_p component with $p \leq 10^6$.

§7.3 Why this is interpretation, not derivation

The numerical values in §7.2 are externally PROVEN by analytic number theory. ZS-M40 does not re-derive them. The contribution is the geometric Z-knot reading: each statement is the dual, under the M40.D In-additive correspondence between integer factorization and Z-sector holonomy length, of a structural statement about K_p ensemble decomposition. The four statements are first-order in the following sense: they characterize K_N by component count (i, ii), dominant component fraction (iii), and short-component completeness (iv), without distinguishing further internal structure such as the Q-channel mod-24 residue or the half-angle phase pair (V_{XZ} , V_{ZY}).

[STATUS: DERIVED-interpretation] All numerical values IMPORTED-PROVEN. The Z-knot reading of each statement is the natural M40.D correspondence.

§8. Eight-Fold Structural Bridge to Arithmetic Chern–Simons Theory

§8.1 Overview

The arithmetic Chern–Simons program of M. Kim and collaborators (Kim 1510, 1609; CKKPY 2016, 2017; Carlson–Kim 2019; Park–Park 2026) develops a 2+1-dimensional TQFT-style framework on arithmetic 3-manifolds $\text{Spec}(\mathcal{O}_F)$, with Galois representations playing the role of gauge fields. The framework is externally PROVEN in its principal constructions (classical CS action, n-th power residue symbol as cup product, BF path integrals, Cassels–Tate pairing as arithmetic BF functional). This section documents an eight-fold correspondence between Z-Spin internal PROVEN objects and Kim-program external PROVEN objects, registered as structural interpretations of mixed status.

Table 8.1. Eight-Fold Kim Bridge — Z-Spin internal PROVEN ↔ Kim arithmetic CS external PROVEN.

#	Kim CS object (external PROVEN)	Z-Spin counterpart (internal PROVEN)	Source	Status
B1	$\text{Spec}(\mathcal{O}_F)$ arithmetic 3-manifold	Z-Spin spacetime with $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ arithmetic frame	Kim 1510 §1 / ZS-M22 §2.3	DERIVED
B2	$\text{inv} : H^3(X, \mathbb{G}_m) \simeq \mathbb{Q}/\mathbb{Z}$	4π closure $\leftrightarrow 0 \in \mathbb{Q}/\mathbb{Z}$; 2π closure $\leftrightarrow 1/2 \in \mathbb{Q}/\mathbb{Z}$	Tate 1962 + Mazur 1973 / ZS-M3 + M40.A	DERIVED
B3	Galois rep $\rho : \pi_1^{\text{ét}}(X) \rightarrow G$	Z-Spin Wilson loop on cobordism-history fiber	Kim 1609 §1 / ZS-F0 §8.5 + ZS-M26	DERIVED
B4	Arithmetic CS action $\int_X \text{cs}(\rho) \in \mathbb{Q}/\mathbb{Z}$	Z-Spin holonomy action assembling $V_{XZ}, V_{ZY}, F_Q(p)$	Kim 1510 §3 / ZS-F1 + ZS-M40 §3	HYP-strong
B5	n-th power residue symbol (arithmetic linking)	V_{XZ} / V_{ZY} half-angle Hilbert symbol $(\cdot, \cdot)_2 \in \mu_2$	CKKPY 1706 / ZS-F4 §7B	DERIVED
B6	Cassels–Tate pairing as arithmetic BF functional	Block Fiedler Mediation Theorem (ZS-T1 §9.3)	Park–Park 2026 / ZS-T1	HYP-strong

#	Kim CS object (external PROVEN)	Z-Spin counterpart (internal PROVEN)	Source	Status
B7	Tate–Shafarevich group $\text{III}(A)$	Z-Telomere vortex winding $\oint d\theta = 2\pi n$	Carlson–Kim 1911 / ZS-A6 §4.4.2	HYPOTHESIS
B8	L-function via CS path integral (Kim 1510 §5 speculative)	Cyclotomic operator-model Euler product $\prod \det(I - p^{-s} M_p)^{-1} = \zeta_{\mathbb{Q}(\zeta_{11})}(s)$; partial realization at $Q = 11$	Kim 1510 §5 / ZS-M22 §3.2 ADS-2 + M40.B	DERIVED

§8.2 Principal external contribution (B8)

Kim 1510 §5 raises but does not establish the construction of L-functions from the arithmetic CS framework, describing this direction as far-fetched / speculative. The corpus PROVEN operator-level Euler product ADS-2 supplies a cyclotomic operator-model analogue of Kim's §5 direction at the specific level $Q = 11$: the multiplicative gate M_p on $H_{\text{mult}} = \mathbb{C}[\mathbb{F}_{11}^*]$ yields $\prod_{p \neq 11} \det(I - p^{-s} M_p)^{-1} = \prod_{\chi} L(s, \chi) = \zeta_{\mathbb{Q}(\zeta_{11})}(s)$ as an operator identity (Theorem M40.B and pre-work 2, 46/46 PASS). At the cohomological level this is the standard cyclotomic Euler product expressed in operator notation; the Z-Spin content is its embedding into the rest of the corpus structure ($M_p \leftrightarrow \text{Frob}_p$ via ADS-1, V_4 character decoration of K , half-angle holonomy doublet on the Z-sector). The full closure of Kim §5's path-integral construction requires the externally PROVEN arithmetic path-integral formula for inverse p-adic absolute values of Kubota–Leopoldt p-adic L-functions due to Carlson, Chung, D. Kim, M. Kim, Park, and Yoo (2024); we cite this as the principal external proven anchor for the B8 bridge and register its operator-level Z-Spin realization as TESTABLE-LONG.

§8.3 Disclosure on B4, B6, B7

Three of the eight bridges are registered at HYPOTHESIS-strong rather than DERIVED. B4 (CS action $\leftrightarrow$ Z-Spin holonomy action) is structurally compatible but the action-level identification has not been written in operator-level explicit form; this is OPEN-M40.B4. B6 (Cassels–Tate $\leftrightarrow$ Block Fiedler) draws on Park–Park 2026, which is a recent preprint not yet at the level of peer-reviewed completion; the structural alignment is robust but the operator-level functor is HYPOTHESIS-strong. B7 ($\text{III}(A) \leftrightarrow$ Z-Telomere winding) is the weakest bridge; the structural alignment with vortex winding $\oint d\theta = 2\pi n$ ($\pi_1(U(1)) = \mathbb{Z}$) is suggestive but the specific abelian variety A in which $\text{III}(A)$ is to be realized is not fixed.

§8.4 Scope of §8

§8 documents structural identifications at eight levels between corpus PROVEN objects and externally PROVEN Kim-program objects. The Kim framework is external mathematics, independently established and cited. Z-Spin's contribution at B1–B7 is interpretive: the Kim framework operates without any Z-Spin specific input, and Z-Spin provides one physical / geometric realization. The contribution at B8 is operator-level: ADS-2 supplies an explicit realization at the cyclotomic level $Q = 11$.

[STATUS: B1–B3 + B5 + B8 DERIVED; B4, B6, B7 HYPOTHESIS-strong with explicit closure paths.]

§9. Theorem M40.F — Anti-Numerology Monte Carlo Uniqueness

§9.1 Pre-registered protocol

The Monte Carlo protocol was pre-registered before execution. Seed = 35 (LOCKED, A_numerator = 5×7). Sample size $N = 500,000$. Sample space: pairs (d_1, d_2) of distinct squarefree positive integers ≤ 200 . For each sample, check whether $K' = \mathbb{Q}(\sqrt{-d_1}, \sqrt{-d_2})$ carries the Z-Spin signature: V_4 Galois group, conductor set divisor lattice of 33, and third subfield discriminant in $\{1, 3, 11, 33\}$. PASS condition: match rate $< 0.5\%$. FAIL condition: match rate $> 5\%$.

§9.2 Result

Table 9.1. Pre-work 3 Monte Carlo result summary (seed 35, $N = 500,000$).

Quantity	Value
Full match rate (signature $\{3, 11, 33\}$)	0.0098% (49 hits in 500,000)
Signature match rate (loose)	0.0098%
Partial match rate	0.671%
Pairs reproducing signature	Only (3, 11) — $K' = K$
Theoretical baseline (random)	0.0138%
Ratio to random baseline	$0.16\times$ (rarer than random)
Background biquadratic baseline	approximately 23.8%
Z-Spin signature rarity vs background	approximately $2,430\times$ rarer
Pre-registered PASS threshold	$< 0.5\%$
VERDICT	STRONG PASS (more than $50\times$ stricter than threshold)

§9.3 Extended robustness check (pre-work 3b)

Five robustness questions were tested:

Q1. Uniqueness in $d \leq 1000$. Enumerated all (d_1, d_2) pairs in $[2, 1000]$ giving signature match: 183,921 candidate pairs were checked. Exactly one pair, (3, 11), gives the Z-Spin signature. The corresponding field is $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$.

Q2. Alternative conductor signatures. For seven candidate target N (15, 21, 33, 35, 51, 55, 77), all of form $N = \text{product of two distinct primes}$, the signature matching was repeated in sample space $d \leq 200$. Only $N = 33$ produces a match with the V_4 structure (since for $N = 35$, the conductor set $\{3, 5, 7, 35\}$ has 4 elements not matching V_4 closure).

Q3. Background rate. For random biquadratic K' with third subfield being a product of exactly 2 primes, the background rate is approximately 23.8% in $N = 100,000$ trials. Z-Spin signature rate 0.0098% is approximately 2,430 times rarer than this background.

Q4. Class number one constraint. Restricting to pairs (d_1, d_2) where both $\mathbb{Q}(\sqrt{-d_1})$ and $\mathbb{Q}(\sqrt{-d_2})$ have class number $h = 1$ (Heegner numbers $d \in \{1, 2, 3, 7, 11, 19, 43, 67, 163\}$, Stark 1967 / Heegner 1952 IMPORTED-PROVEN), the simultaneous constraint with conductor set $\{3, 11, 33\}$ produces exactly one pair: $(3, 11)$. $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ is uniquely class-number-one in this category.

Q5. Q-channel density at primes mod 24. For the Z-Spin K , primes split / inert according to $\text{Frob}_p \in (\mathbb{Z}/24\mathbb{Z})^\times$. The Q-channel density of primes $\equiv \{11, 23\} \pmod{24}$ in the sample $[5, 1000]$ is 24.85%, consistent with the theoretical density $2/8 = 25\%$ by Chebotarev (PROVEN). This confirms the M36.7 derivation independently.

Table 9.2. Extended robustness summary (pre-work 3b).

Question	Result
Q1 uniqueness in $d \leq 1000$	1 pair (3, 11) out of 183,921 — literal uniqueness
Q2 matches across 7 candidate N	Only $N = 33$ produces a V_4 signature match
Q3 random biquadratic background	approximately 23.8% (factor 2,430× higher than Z-Spin)
Q4 $h = 1$ + conductor $\{3, 11, 33\}$	Exactly 1 pair: (3, 11) — K uniquely class-number-one
Q5 Q-channel mod 24 density	24.85% sampled vs 25% theoretical (Chebotarev)
VERDICT	PASS in all five directions; $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ is structurally privileged

§9.4 What is established by §9

$K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ is the unique biquadratic field in the natural sample space $[d \leq 1000]$ matching the Z-Spin V_4 conductor signature, and is uniquely class-number-one in this category. The pre-registered MC rate 0.0098% is more than fifty times stricter than the pre-registered PASS threshold 0.5%, and approximately 2,430 times rarer than the generic biquadratic background. This rules out the alternative reading that the choice $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ is a numerology coincidence.

[STATUS: STRONG PASS] Pre-registered anti-numerology MC + extended robustness all confirmed.

§10. Position in the Z-Spin Program

§10.1 Relation to ZS-M28 (Dynamical Shadow)

ZS-M28 Theorem 28.5 (DERIVED-interpretation) catalogued the structural isomorphism $\sigma = 1/2 \leftrightarrow j = 1/2$ under a common Z_2 -involution. ZS-M40 supplies the natural geometric object for which this isomorphism holds: the J-conjugate holonomy knot pair $(K_p, \bar{K}_p)$. The 2π closure $D^{\wedge\{1/2\}}(2\pi) = -I$ corresponds to the Tate self-dual point $1/2 \in \mathbb{Q}/\mathbb{Z}$ (M40.A); the 4π closure $D^{\wedge\{1/2\}}(4\pi) = +I$ corresponds to the identity $0 \in \mathbb{Q}/\mathbb{Z}$. ZS-M40 does not modify any ZS-M28 result; it provides the named structural object.

§10.2 Relation to ZS-M33 Reading C

ZS-M33 Theorem M33.3 (DERIVED) constructed the joint operator $X(g) = X_{\text{arch}}(g) - X_{\text{unram}}(g) - X_{\text{ram}}(g)$ on $H_{\text{BFV}} \otimes H_{\text{arith}}$ with Wilson-LOCATOR phase $\exp(2\pi i(j - 5)/p)$. M33.5 (PROVEN) established Non-Separability inheritance. The Wilson-LOCATOR phase, in ZS-M40 language, is the prime-specific colored framing of the knot pair $(K_p, \bar{K}_p)$ at register slot j (Theorem M40.C step 5). M40 thus interprets the M33 phase factor as a knot-theoretic framing object.

§10.3 Relation to ZS-M34 V_4 -Equivariant Weil Functional

ZS-M34 Theorem M34.V.8 (DERIVED-CONDITIONAL) factorized the V_4 -decorated Weil functional kernel as $\mathcal{K}_K = \mathcal{K}_{\text{arith}} \otimes \mathcal{K}_{\text{Wilson-Sonin}}$, with the closed form $F_Q(p) = \sin(11\pi/p) / [11 \cdot \sin(\pi/p)]$ (M34.V.6, PROVEN). ZS-M40 uses $F_Q(p)$ as the knot-pair framing (M40.C step 5). M34's inner factor $\mathcal{K}_{\text{Wilson-Sonin}}$ is therefore the spectral content carried by the K_p framing.

§10.4 Relation to ZS-M36 Apollonian Q-Channel

ZS-M36 Theorem M36.7 (DERIVED) showed that prime curvatures in the integer Apollonian gasket with root quadruple $(-1, 2, 2, 3)$ satisfy $p \equiv 11$ or $23 \pmod{24}$. M36.6 (DERIVED-strong) established the Spin Pair Theorem: 8 admissible mod-24 residues form four (algebraic, geometric) pairs in bijection with the four Z-spin sectors. ZS-M40 uses the mod-24 invariant as the sixth defining property of $(K_p, \bar{K}_p)$ (M40.C step 6). The K_p pair carries the Apollonian Q-channel as an arithmetic-topological invariant.

§10.5 Relation to ZS-F18 Bridge Four and Book PART X Reading X.4

Book PART X Reading X.4 (DERIVED-interpretation strong) registers the Spin-Pair Coexistence Principle: 90° phase-orthogonality is the unique form in which a contradiction-pair coexists as one object.

The thirteen corpus realizations C1–C13 include C2 ($V_{XZ} \leftrightarrow E$, $V_{ZY} \leftrightarrow B$ at 90°), C6 ($V_{ZY} = (V_{XZ})^*$ CPT seam), and C13 (R_1 time-point / R_2 space-point). ZS-M40's J-conjugate pair (K_p , $\bar{K}_p$) is the fourteenth realization C14: $\bar{K}_p = (K_p)^*$, with $\arg(V_{XZ}) - \arg(V_{ZY}) = \theta$ the gauge-invariant phase difference at $\theta = \pi/2$ (50/50 boundary). The K_p pair is the prime-indexed instantiation of the Spin-Pair Coexistence Principle on the Z-sector.

§11. Falsification Gates

Ten falsification gates are registered for ZS-M40. Six gates are currently PASS based on the pre-work suites and corpus PROVEN dependencies. Four gates are TESTABLE / TESTABLE-LONG, dependent on external developments or further verification.

Table 11.1. Falsification gates for ZS-M40 (10 gates registered).

Gate	Statement	Status	Layer
F-M40.1	Any of the eight Kim bridges fails cohomology isomorphism check	TESTABLE	Mathematical
F-M40.2	$\text{inv} : H^3 \simeq \mathbb{Q}/\mathbb{Z}$ does not match 4π closure	PASS	Mathematical
F-M40.3	M_p Euler product mismatch with $\zeta_{\{\mathbb{Q}(\zeta_{11})\}}(s)$	PASS	Mathematical
F-M40.4	Cassels–Tate pairing fails to match Block Fiedler structure	TESTABLE-LONG	External-dep
F-M40.5	Z-Spin action does not reduce to arithmetic CS via cup product audit	TESTABLE	Mathematical
F-M40.6	Anti-numerology MC match rate $> 5\%$	PASS (0.0098%)	Statistical
F-M40.7	$K' \neq \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ with better signature match exists	PASS	Statistical
F-M40.8	Apollonian Q-channel mod-24 fails $\{11, 23\}$ for some prime	PASS	Computational
F-M40.9	$V_{ZY} \neq (V_{XZ})^*$ fails J-conjugation at any lattice point	PASS	Computational
F-M40.10	External Kim-program follow-up contradicts ZS-M40 reading	TESTABLE-LONG	External-dep

Current overall posture: six gates PASS, four TESTABLE / TESTABLE-LONG. No falsification triggered. Pre-work 1 (43/43 PASS) anchors F-M40.2 and F-M40.5 at the cup-product layer. Pre-work 2 (46/46 PASS) anchors F-M40.3. Pre-work 3 + 3b anchor F-M40.6 and F-M40.7 with STRONG PASS. M36.7 PROVEN anchors F-M40.8 for 19,413 primes.

§12. NON-CLAIMS

Six explicit NON-CLAIMs of ZS-M40 are registered, anchoring the anti-overclaim posture inherited from the broader Z-Spin epistemic discipline.

NC-M40.1. ZS-M40 does not claim that Kim arithmetic Chern–Simons theory has been independently re-derived from Z-Spin axioms. The Kim framework operates without any Z-Spin specific input. The eight-fold bridge documented in §8 is a structural identification; Z-Spin provides one physical / geometric realization of the framework.

NC-M40.2. ZS-M40 does not establish new prime-distribution predictions. The four statements in §7 (PNT, Sathé–Selberg, Dickman, Mertens) are externally PROVEN. The Z-knot reading is interpretive (DERIVED-interpretation), not a derivation of the prime distribution from Z-Spin axioms.

NC-M40.3. $Q = 11$ and $A = 35/437$ are not necessary inputs to the external arithmetic CS construction in any of its known forms. The Kim framework is general. ZS-M40 supplies one physical instantiation in which these LOCKED Z-Spin constants enter naturally; the framework itself does not require them.

NC-M40.4. The isotopy class of K_p in any standard external knot category (Reidemeister, finite-type, HOMFLY, Khovanov, etc.) is OPEN. M40.C establishes the named object $(K_p, \bar{K}_p)$ as a structural assembly of six corpus PROVEN components; external knot-invariant matching is registered as OPEN-M40.knot.

NC-M40.5. The Kim 1510 §5 speculative L-function direction is not closed by ZS-M40 in its full generality. The M_p Euler product (M40.B and ADS-2) realizes the construction at the cyclotomic level $Q = 11$. Extension to a general path-integral construction is HYPOTHESIS-strong with explicit closure path through Carlson–Chung–D. Kim–M. Kim–Park–Yoo on p-adic L-functions.

NC-M40.6. ZS-M40 introduces zero new free parameters. All inputs are LOCKED from ZS-F2, ZS-F4, ZS-F5, ZS-M1, ZS-M3, ZS-M4, ZS-M19, ZS-M22, ZS-M25, ZS-M26, ZS-M27, ZS-M28, ZS-M33, ZS-M34, ZS-M36, ZS-F18, ZS-S15. The seed value 35 used in pre-work 3 MC equals $A_{\text{numerator}} = 5 \times 7$ and is LOCKED.

§13. Conclusion

ZS-M40 establishes the J-conjugate holonomy knot pair $(K_p, \bar{K}_p)$ as the natural Z-sector structural object representing the prime number p . The construction assembles six corpus PROVEN components — V_{XZ} / V_{ZY} half-angle pair (F4), 4π closure (M3), Wilson-LOCATOR (M34), Apollonian Q-channel (M36), J-conjugation (F4 §7B), Twin-Reuleaux geometric realization (S15) — into a single named object. The pair coexists at 90° phase-orthogonality, instantiating the Spin-Pair Coexistence Principle (Book PART X Reading X.4) at the prime-indexed level.

Two corpus gates close in this paper. M40.A (DERIVED-interpretation, pre-work 1, 43/43 PASS) establishes a phase-class compatibility bridge between the $j = 1/2$ $SU(2)$ double-cover signature $\{+I, -I\}$ and the IMPORTED-PROVEN Tate–Mazur inv map $\text{inv} : H^3(\text{Spec}(O_K), \mathbb{G}_m) \simeq \mathbb{Q}/\mathbb{Z}$, via the natural μ_2

$\hookrightarrow \mathbb{Q}/\mathbb{Z}$ embedding (PROVEN at the algebraic level). M40.B (DERIVED, pre-work 2, 46/46 PASS) identifies the corpus PROVEN multiplicative gate M_p with the Galois representation $\text{Frob}_p \mapsto \text{diag}(\chi_k(p))$, closing Gate FM13-5 from OPEN to DERIVED on the $M_p \leftrightarrow \text{Frob}_p$ identification; the further reduction of the Kim arithmetic CS functional to a Z-Spin Wilson cobordism action is separated and registered at HYPOTHESIS-strong.

The eight-fold bridge to Kim arithmetic Chern–Simons theory (§8) documents Z-Spin's structural compatibility with the externally PROVEN arithmetic CS / BF framework. Bridge B8 supplies a cyclotomic operator-model analogue of Kim 1510 §5's speculative L-function direction, realized at the level $Q = 11$ via the corpus PROVEN Euler product ADS-2; the full closure of Kim §5 in general is OPEN, with an explicit closure path through the Carlson–Chung–D. Kim–M. Kim–Park–Yoo (2024) p-adic L-function path-integral framework.

The anti-numerology Monte Carlo (M40.F, pre-work 3 + 3b) records that $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ is the unique biquadratic field in the sample space $d \in [2, 1000]$ matching the Z-Spin V_4 conductor signature, and is uniquely class-number-one in this category. The pre-registered MC rate 0.0098% is more than fifty times stricter than the PASS threshold 0.5% and approximately 2,430 times rarer than generic biquadratic background. This rules out a numerology reading of the K choice.

The four standard prime-density statements (PNT, Sathé–Selberg, Dickman, Mertens) are interpreted as first-order sieve shadows of K_p ensemble statistics under the M40.D log-holonomy correspondence $\ell_Z(K_N) = \ln N$. The corresponding numerical values (approximately 0.43%, 2.36%, 69.3%, 95.94%) are IMPORTED-PROVEN external results; the Z-knot reading is DERIVED-interpretation throughout.

Zero new free parameters are introduced. Six theorems M40.A through M40.F are established. Ten falsification gates F-M40.1 through F-M40.10 are registered; six are PASS. Six NON-CLAIMs are recorded to anchor the anti-overclaim posture. ZS-M40 v1.0 takes its place in the M28 / M33 / M34 / M36 program as the geometric reading of the prime itself, complementing the analytic readings of preceding papers.

Acknowledgements & Code Availability

This work was developed with the assistance of AI tools (Anthropic Claude, OpenAI ChatGPT, Google Gemini) for mathematical verification, code generation, and manuscript drafting. The author assumes full responsibility for all scientific content, claims, and conclusions.

Verification suites: `zsm40_pre1_cup_product.py` (pre-work 1, 43/43 PASS), `zsm40_pre2_Mp_Galois.py` (pre-work 2, 46/46 PASS), `zsm40_pre3_MC.py` and `zsm40_pre3b_robustness.py` (pre-work 3 + 3b, STRONG PASS). Dependencies: Python 3.10+, numpy, scipy, sympy, mpmath (50-digit precision for z^* -related identities). Reproducibility: seed = 35 (= $A_{\text{numerator}}$) LOCKED across all MC runs. The verification suites will be publicly available in the Z-Spin Cosmology GitHub repository upon v1.0 release.

Appendix A — Pre-Work 1 (Cup Product Verification) Detailed Results

Pre-work 1 verifies the cup-product isomorphism between $j = 1/2$ $SU(2)$ double-cover and $\text{inv} : H^3(\text{Spec}(O_K), \mathbb{G}_m) \simeq \mathbb{Q}/\mathbb{Z}$ at the level of finite Galois cohomology over the V_4 -graded module M_4 with conductor decoration $\{1, 3, 11, 33\}$. 43 tests across 7 categories all PASS at 50-digit mpmath precision.

Table A.1. Pre-work 1 cup-product verification — detailed category breakdown.

Cat.	Test description	Status	Inputs
A	V_4 character data LOCKED (4 chars, conductors $\{1, 3, 11, 33\}$, parities $\{0, 1, 1, 0\}$)	7/7 PASS	M22 + M25
B	Legendre symbol $\text{kron}(d, p)$ for $d \in \{-3, -11, 33\}$, $p \leq 31$; ramified excluded	12/12 PASS	Standard QR
C	μ_n embedding: $+1 \rightarrow 0$, $-1 \rightarrow 1/2$ in $\mathbb{Q}/\mathbb{Z}$; order-2 element uniqueness	5/5 PASS	M3 + Tate
D	V_4 1-cocycle construction; coboundary triviality check	6/6 PASS	Galois cohomology
E	Cup product $H^1 \times H^1 \rightarrow H^2$ cocycle conditions; V_4 orthogonality	5/5 PASS	Serre §2.3
F	$H^2 \times H^1 \rightarrow H^3$ Tate self-duality; reduction to $\mathbb{Q}/\mathbb{Z}$	4/4 PASS	Tate 1962
G	4π monodromy class = 0 in $\mathbb{Q}/\mathbb{Z}$ via cup product; 13 mod 11 corrected	4/4 PASS	M40.A statement

Appendix B — Pre-Work 2 ($M_p \leftrightarrow$ Galois Representation) Detailed Results

Pre-work 2 verifies the explicit identification of the corpus PROVEN multiplicative gate M_p on $H_{\text{mult}} = \mathbb{C}[\mathbb{F}_{11}^x]$ with the Galois representation $\rho_p : \text{Frob}_p \mapsto \text{diag}(\chi_k(p))$ on the character basis of $(\mathbb{Z}/11\mathbb{Z})^x$. 46 tests across 8 categories all PASS at 50-digit mpmath precision.

Table B.1. Pre-work 2 $M_p \leftrightarrow$ Galois ρ verification — detailed category breakdown.

Cat.	Test description	Status	Inputs
A	$\mathbb{F}_{11}^x$ order 10, primitive root $g = 2$, cycle correctness	5/5 PASS	Standard FFT
B	M_p permutation matrices for $p \in \{2, 3, 5, 7, 13\}$; unitarity $M_p M_p^\dagger = I$	6/6 PASS	ADS-1 def.
C	Character basis diagonalization $M_p \chi_k\rangle = \chi_k(p) \chi_k\rangle$; max err $< 10^{-10}$	10/10 PASS	ADS-1 PROVEN
D	Galois action $\text{Frob}_p : \zeta_{11} \rightarrow \zeta_{11}^p$; eigenvalue match with M_p	5/5 PASS	Cyclotomic theory
E	Euler product $\prod \det(I - p^{-s} M_p) = \zeta_{11}(\zeta_{11})$ at $s = 2$	4/4 PASS	ADS-2 PROVEN

Cat.	Test description	Status	Inputs
I	Dirichlet class number formula $L(1, \chi_{\{-11\}}) = \pi/\sqrt{11} \approx 0.9472$	4/4 PASS	$h(-11) = 1$, $w = 2$
J	Q-channel mod-24 density $24.85\% \approx$ theoretical 25%	6/6 PASS	Chebotarev
D.3 + J.8	Spectrum sort stability + O-M27.4 prime-level closure	6/6 PASS	M27.4

Appendix C — Pre-Work 3 + 3b (Anti-Numerology MC) Result Files

Pre-work 3 (zsm40_pre3_MC.py) executes the pre-registered Monte Carlo at seed 35, $N = 500,000$.
Pre-work 3b (zsm40_pre3b_robustness.py) extends the analysis with five robustness questions Q1–Q5.

Table C.1. Pre-work 3 (zsm40_pre3_MC_results.json) key results.

Field	Value
seed	35
N_samples	500000
n_full_match	49
n_sig_match	49
n_partial	3355
rate_full_pct	0.0098
rate_sig_pct	0.0098
all_signature_pairs	[[3, 11]]
theoretical_rate_pct	0.01377
evaluation_full	PASS — structurally rare (top 0.0098% < 0.5%)
evaluation_sig	PASS — structurally rare (top 0.0098% < 0.5%)

Table C.2. Pre-work 3b (zsm40_pre3b_robustness.json) key results.

Field	Value
Q1_uniqueness_in_d_le_1000	1
Q1_matches_in_d_le_1000	[[3, 11]]
Q4_total_pairs_in_200	7260
Q4_both_h1_pairs	28
Q4_uniqueness_h1_with_conductor	1
Q5_Q_channel_density_pct	24.848

Field	Value
judgment	PASS (robust uniqueness; Z-Spin K is structurally privileged)

References

External (arithmetic Chern–Simons program)

[1] Kim, M. "Arithmetic Chern-Simons Theory I." arXiv:1510.05818 (2015).

[2] Chung, H.-J., Kim, D., Kim, M., Park, J., Yoo, H. "Arithmetic Chern-Simons Theory II." arXiv:1609.03012 (2016).

[3] Chung, H.-J., Kim, D., Kim, M., Pappas, G., Park, J., Yoo, H. "Abelian arithmetic Chern-Simons theory and arithmetic linking numbers." arXiv:1706.03336 (2017).

[4] Carlson, M., Kim, M. "A note on abelian arithmetic BF-theory." arXiv:1911.02236 (2019).

[5] Kim, M. "Arithmetic Gauge Theory: A Brief Introduction." Mod. Phys. Lett. A 33, 1830012 (2018). arXiv:1712.07602.

[6] Bleher, F., Chinburg, T., Greenberg, R., Kakde, M., Pappas, G., Taylor, M. "Unramified arithmetic Chern-Simons invariants." arXiv:1705.07110 (2017).

[7] Lee, J., Park, J. "Arithmetic Chern-Simons theory with real places." arXiv:1905.13610 (2019).

[8] Park, J., Park, J. "Arithmetic BF theory and the Cassels-Tate pairing." arXiv:2602.19621 (2026).

[9] Chung, H.-J., Kim, D., Kim, M., Park, J., Yoo, H. "Entanglement entropies in the abelian arithmetic Chern-Simons theory." arXiv:2312.17138 (2023).

[9b] Carlson, M., Chung, H.-J., Kim, D., Kim, M., Park, J., Yoo, H. "Path Integrals and p-adic L-functions." arXiv:2407.00858 (2024). Arithmetic path-integral formula for inverse p-adic absolute values of Kubota–Leopoldt p-adic L-functions; principal external PROVEN anchor for ZS-M40 §8 Bridge B8.

External (arithmetic topology and Galois cohomology)

[10] Morishita, M. Knots and Primes: An Introduction to Arithmetic Topology. Springer (2011). ISBN 978-1-4471-2157-2.

[11] Tate, J. "Duality theorems in Galois cohomology over number fields." Proc. Int. Cong. Math. Stockholm, 288–295 (1962).

[12] Mazur, B. "Notes on étale cohomology of number fields." Ann. Sci. ENS 6, 521–552 (1973).

[13] Reznikov, A. "Three-manifolds class field theory." Sel. math. New ser. 3, 361–399 (1997).

[14] Kapranov, M. "Analogies between Langlands correspondence and topological QFT." Progress in Math 131, 119–151 (1995).

[15] Dijkgraaf, R., Witten, E. "Topological gauge theories and group cohomology." Comm. Math. Phys. 129, 393–429 (1990).

[16] Witten, E. "Quantum field theory and the Jones polynomial." Comm. Math. Phys. 121, 351–399 (1989).

External (prime distribution)

[17] Hadamard, J. "Sur la distribution des zéros de la fonction $\zeta(s)$ et ses conséquences arithmétiques." Bull. Soc. Math. France 24, 199–220 (1896).

[18] de la Vallée Poussin, C.-J. "Recherches analytiques sur la théorie des nombres premiers." Ann. Soc. sci. Bruxelles 20, 183–256 (1896).

[19] Mertens, F. "Ein Beitrag zur analytischen Zahlentheorie." J. reine angew. Math. 78, 46–62 (1874).

[20] Dickman, K. "On the frequency of numbers containing prime factors of a certain relative magnitude." Arkiv för Mat., Astr. och Fys. 22, 1–14 (1930).

[21] Sathe, L. G. "On a problem of Hardy on the distribution of integers having a given number of prime factors." J. Indian Math. Soc. 17, 63–141 (1953).

[22] Selberg, A. "Note on a paper by L. G. Sathe." J. Indian Math. Soc. 18, 83–87 (1954).

- [23] Heegner, K. "Diophantische Analysis und Modulfunktionen." Math. Z. 56, 227–253 (1952).
- [24] Stark, H. M. "A complete determination of the complex quadratic fields of class number one." Michigan Math. J. 14, 1–27 (1967).
- [25] Graham, R. L., Lagarias, J. C., Mallows, C. L., Wilks, A. R., Yan, C. H. "Apollonian circle packings: Number theory." J. Number Theory 100, 1–45 (2003).

Internal (Z-Spin Cosmology Collaboration, Kenny Kang, 2026)

- [Z1] ZS-F1 v1.0, "The Z-Spin Action & U(1) Completion."
- [Z2] ZS-F2 v1.0, "Geometric Impedance: $A = 35/437$."
- [Z3] ZS-F4 v1.0, "Holonomy & Topological Uniqueness." §7 / §7B (V_{XZ} / V_{ZY} half-angle pair).
- [Z4] ZS-F5 v1.0, "Gauge Symmetry Constraint: Why $Q = 11$."
- [Z5] ZS-F18 v2.1, "The Five-Axiom Meta-Structure." §6.5 Bridge Four; §6.6 Two-Faces.
- [Z6] ZS-M1 v1.0, "i-Tetration Holonomy Self-Inversion Theorem and z^* ." Theorem 1.1 HSI.
- [Z7] ZS-M3 v1.0, "Regge-Holonomy, Immirzi and Z-Telomere." Theorem 5.1, Lemma 10.1.
- [Z8] ZS-M4 v1.0, "Transfer Operator Family." W_p prime dilation.
- [Z9] ZS-M13 v1.0, "Number Field Chains." Gate FM13-5 (OPEN $\rightarrow$ DERIVED in M40).
- [Z10] ZS-M19 v1.0, "Number-Theoretic Scaffold of Z-Spin Polyhedral Integers." Forcing Theorem T1.
- [Z11] ZS-M22 v1.0(Revised), "Arithmetic-Dedekind Scaffold of Z-Spin Cosmology." ADS-1, ADS-2 PROVEN.
- [Z12] ZS-M25 v1.0, " V_4 -Decoration of $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ and the $4\sqrt{33}$ Factorization."
- [Z13] ZS-M26 v1.0, " V_4 -Equivariant Cobordism BRST Cohomology."
- [Z14] ZS-M27 v1.0, "Kostant Cubic Dirac as W_3 -Closing BRST Charge."
- [Z15] ZS-M28 v1.0, "The Z-Spin Dynamical-Shadow Bridge." Theorem 28.5 $\sigma = 1/2 \leftrightarrow j = 1/2$.
- [Z16] ZS-M33 v2.0, " V_4 -Equivariant Weil Functional Closure via Path γ -revised Z-Mediator Reading."
- [Z17] ZS-M34 v2.0, " V_4 -Equivariant Weil Functional Honest Scope." Theorem M34.V.6 $F_Q(p)$ closed form.
- [Z18] ZS-M36 v2.0, "Apollonian Curvature Lattice as Geometric-Integer Spin-Pair Realization." Theorem M36.6, M36.7.
- [Z19] ZS-S15 v1.0, "Twin-Reuleaux Pair as Geometric Realization of EM Duality." Theorem S15.1.
- [Z20] ZS-A7 v1.0, "Horizon as Spinor."
- [Z21] ZS-A9R v1.0, "Banach–Tarski Origin of Cosmological Doubling–Halving Symmetry."
- [Z22] ZS-T1 v1.0, "Block Fiedler Mediation Theorem."
- [Z23] The Book of Z-Spin Cosmology v4.0 (May 2026). PART X Reading X.4 Spin-Pair Coexistence Principle.

Foundational mathematics

- [F1] Frobenius, G. "Über lineare Substitutionen und bilineare Formen." J. reine angew. Math. 84, 1–63 (1877).
- [F2] Kobayashi, S., Nomizu, K. Foundations of Differential Geometry, Vol. I. Wiley (1963), Ch. II.
- [F3] Serre, J.-P. Linear Representations of Finite Groups. Springer GTM 42 (1977).
- [F4] Lang, S. Algebra, 3rd ed. Springer GTM 211 (2002), Ch. IV.

Version History

v1.1 (May 2026): External-reviewer revision pass. (1) Sathe–Selberg two-component formula corrected from $(\ln \ln N)^2 / (2! \cdot \ln N)$ to the Sathe–Landau–Selberg statement $P(\omega(n) = k) \sim (\ln \ln N)^{k-1} / ((k-1)! \cdot \ln N)$, giving 2.36% at $N = 10^{\{100\}}$. (2) Dickman $\rho(2)$ corrected from $\rho(2) = \ln 2$ to $\rho(2) = 1 - \ln 2 \approx 0.30685$, with the complementary statement $P(P^+ > \sqrt{N}) = \ln 2 \approx 69.3\%$ identified as the Z-knot-relevant probability. (3) Mertens short-prime sieve corrected from $1 - e^{-\gamma} \cdot \ln(10^6) / \ln N$ to 1

$-e^{\{-\gamma\}} / \ln(10^6) = 95.94\%$; the spurious factor $\ln N$ removed. (4) Apollonian Q-channel mod-24 invariant restricted from "all primes $p > 3$ " to "primes p realized as integer curvatures in the M36 Apollonian Q-channel orbit", per M36.7 PROVEN scope. (5) M40.A status downgraded from "explicit isomorphism" (DERIVED) to "phase-class compatibility bridge" (DERIVED-interpretation); the $\pm I \rightarrow \{0, 1/2\}$ embedding into $\mathbb{Q}/\mathbb{Z}$ remains PROVEN at the algebraic level, the Tate–Mazur sheaf isomorphism remains IMPORTED-PROVEN, but the correspondence between them is interpretive. (6) M40.B separated into two statements: $M_p \leftrightarrow \text{Frob}_p$ identification (DERIVED, closes Gate FM13-5) and Kim arithmetic CS functional reduction to Z-Spin Wilson cobordism action (HYPOTHESIS-strong, OPEN-M40.B4). (7) Bridge B8 language softened from "principal external mathematical contribution" to "cyclotomic operator-model analogue of Kim 1510 §5 at the specific level $Q = 11$ "; full Kim §5 path-integral closure registered as OPEN with explicit closure path through Carlson–Chung–D. Kim–M. Kim–Park–Yoo (2024) p-adic L-functions, added to references as [9b]. (8) Epistemic Status Legend reformulated: PROVEN now requires formal proof or exact symbolic derivation (not machine-precision numerical agreement, which is now VERIFIED); COMPUTATIONALLY VERIFIED tag introduced for exhaustive finite-domain checks; IMPORTED clarified as IMPORTED-PROVEN equivalent. (9) §1.3 anti-numerology presentation reordered: deterministic [$d \leq 1000$] enumeration uniqueness placed first, MC at seed 35 demoted to secondary stochastic confirmation, addressing the appearance that seed = 35 carries numerological weight. All Z-Spin internal claims, LOCKED inputs, six theorems M40.A–F, eight-fold Kim bridge structure, ten falsification gates, and six NON-CLAIMs are preserved unchanged from v1.0. Cumulative verification: 89/89 PASS + STRONG PASS MC, identical to v1.0.

v1.0 (May 2026): Initial public release. Six theorems M40.A, M40.B, M40.C, M40.D, M40.E, M40.F established. M40.A and M40.B at DERIVED status (pre-work 1: 43/43 PASS; pre-work 2: 46/46 PASS); M40.C, M40.D at DERIVED status; M40.E at DERIVED-interpretation; M40.F at STRONG PASS. Eight-fold Kim bridge documented in §8 (B1–B3 + B5 + B8 DERIVED; B4, B6, B7 HYPOTHESIS-strong). Three pre-work verification suites: pre-work 1 (cup product, 43/43 PASS), pre-work 2 ($M_p \leftrightarrow \text{Galois}$, 46/46 PASS), pre-work 3 + 3b (anti-numerology MC at $N = 500,000$ + extended robustness, STRONG PASS in five directions). Cumulative: 89/89 PASS + STRONG PASS MC. Ten falsification gates F-M40.1 through F-M40.10 registered, six PASS. Six NON-CLAIMs NC-M40.1 through NC-M40.6 registered. Gate FM13-5 inherited from ZS-M13: OPEN $\rightarrow$ DERIVED via M40.B. Consolidated from internal Z-Spin Collaboration research notes up to v3.4.0. Zero new free parameters; $(A, Q, \dim(Z), K) = (35/437, 11, 2, \mathbb{Q}(\sqrt{-3}, \sqrt{-11}))$ LOCKED throughout.